



Argon Industria HMI 7C Enclosure for Official Raspberry Pi 7-Inch Screen

Product Datasheet

Product Name	Argon Industria HMI 7C Enclosure for Official Raspberry Pi 7-Inch Screen
Product Model	A25-004-AR-HMI-7C
EAN / UPC	5065019946263
Data Sheet Version	Rev 4 (build v5)
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Argon 40 SKU	ARG-IND-HMI-7C

Overview

Industria enclosure designed to house the official Raspberry Pi 7-inch Touch Display 2. Integrates the RPi display into the Industria industrial form factor.

1. Application

- Raspberry Pi 5 / Raspberry Pi 7-inch Touch Display 2
- Raspberry Pi 5

Product Features

- Aluminum construction with rugged industrial-grade design
- Passive heatsink with optional active cooling module
- Full access to USB, Ethernet, HDMI, and GPIO ports
- VESA, panel, and desktop mounting options
- Compatible with Raspberry Pi HAT modules via GPIO
- M.2 NVMe storage expansion support via PCIe
- Supports Argon Industria Communication modules
- Official Raspberry Pi 7-inch Touch Display 2 integration
- Aluminum construction for durability
- VESA, panel, and desktop mounting compatible
- GPIO access supports Raspberry Pi HATs and Cytron Industrial HAT
- Compatible with Argon Industria Communication modules
- M.2 NVMe expansion board available as separate accessory
- Rugged industrial-grade aluminum design with precision-machined hardware
- Seamless integration into control panels and automation applications
- Industrial HMI (Human-Machine Interface) form factor

2. Product Specifications

- **Category:** AI HMI
- **Weight (kg):** 0.676
- **Dimensions (cm):** 22.4 x 13.6 x 6.4 cm
- **accessible_ports:** USB, Ethernet, HDMI, and GPIO ports
- **communication_modules:** Compatible with Argon Industria Communication modules
- **compatible_display:** Official Raspberry Pi 7-inch Touch Display 2 (720 × 1280 resolution via MIPI DSI)
- **compatible_raspberry_pi:** Raspberry Pi 5
- **compatible_sbc:** Raspberry Pi 5

- **components_sold_separately:** Raspberry Pi 5, Raspberry Pi 7-inch Touch Display 2, active cooling module, M.2 NVMe board, Communication modules
- **cooling:** Passive heatsink (active module available separately)
- **cooling_optional:** Active cooling module
- **cooling_primary:** Passive heatsink
- **design_classification:** Rugged, industrial-grade
- **design_grade:** Industrial-grade, rugged
- **dimensions_unit:** millimeters
- **display_resolution:** 720 × 1280 via MIPI DSI
- **display_type:** Official Raspberry Pi 7-inch Touch Display 2
- **document_type:** Drawing
- **expandability:** Argon Industria Communication modules, M.2 NVMe board
- **expansion_storage:** M.2 NVMe via PCIe 2.0 ×1
- **form_factor:** Industrial HMI (Human-Machine Interface) enclosure
- **gpio_access:** Full 40-pin Raspberry Pi standard header
- **gpio_header:** Full 40-pin Raspberry Pi standard GPIO header access
- **hat_compatibility:** Standard Raspberry Pi HATs; Cytron Industrial HAT
- **hat_support:** Standard Raspberry Pi HAT modules
- **integrated_display:** Official Raspberry Pi 7-inch Touch Display 2
- **intended_applications:** Control panels, smart interfaces, modern industry applications, automation
- **material:** Aluminum
- **mounting_options:** VESA, panel-mount, and desktop mounting
- **mounting_types:** VESA, panel-mount, desktop
- **nvme_expansion_board:** Argon Industria PCIe M.2 Single NVMe Board (sold separately)
- **nvme_support:** M.2 NVMe via PCIe 2.0 ×1
- **optional_active_cooling:** Available separately
- **optional_cooling:** Active cooling module (sold separately)
- **ports_accessible:** USB, Ethernet, HDMI, GPIO
- **precision_hardware:** Precision-machined
- **primary_cooling:** Passive heatsink
- **product_form_factor:** Industrial HMI enclosure
- **product_name:** HMI 7"
- **rpi5_bluetooth:** Bluetooth 5.0 / BLE
- **rpi5_board_dimensions:** 85 mm × 56 mm
- **rpi5_display_interface:** MIPI DSI (used for Touch Display 2 integration), MIPI CSI
- **rpi5_ethernet:** Gigabit Ethernet (1 Gbps)
- **rpi5_gpio:** Standard 40-pin Raspberry Pi GPIO header
- **rpi5_gpu:** VideoCore VII — OpenGL ES 3.1, Vulkan 1.2
- **rpi5_hdmi:** 2× micro-HDMI — supports dual 4Kp60 simultaneous output
- **rpi5_minimum_psu:** 5 V / 3 A (15 W) USB-C
- **rpi5_os_support:** Raspberry Pi OS (64-bit recommended), Ubuntu, and others
- **rpi5_power_button:** Yes — onboard power button
- **rpi5_power_input:** USB-C Power Delivery
- **rpi5_processor:** Broadcom BCM2712, 2.4 GHz quad-core 64-bit Arm Cortex-A76
- **rpi5_ram_options:** 2 GB, 4 GB, 8 GB, or 16 GB LPDDR4X-4267
- **rpi5_recommended_psu:** 5 V / 5 A (27 W) USB-C PD
- **rpi5_rtc:** Yes — onboard RTC with battery connector
- **rpi5_storage:** microSD (UHS-I)
- **rpi5_usb_2:** 2× USB 2.0
- **rpi5_usb_3:** 2× USB 3.0 (5 Gbps)
- **rpi5_wifi:** Dual-band 802.11ac (2.4 GHz + 5 GHz)
- **scale:** 1:1
- **sku:** ARG-IND-HMI-7C

- **specific_hat_compatibility:** Cytron Industrial HAT
- **storage_expansion:** M.2 NVMe via PCIe 2.0 ×1 (expansion board sold separately)

3. Mechanical Information and Functional Diagram

Diagram not available for this revision — refer to the product manual.

4. Reference Design

- (Electronics Schematics)

5. Product Compliance Information

- This part is compliant with EU 2002/95/EC RoHS directive (Restrictions on the Use of Certain Hazardous Substances in Electrical and Electronic Equipment)

6. Contact Information

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